



Practical Medical Microbiology

Safety rules and lab. Equipment

Lab. 1

Third stage- College of Dentistry

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Introduction:

- ❖ Laboratories often involve significant risks, making accident prevention essential.
- ❖ **Key risk factors** include exposure to high voltage equipment, extreme pressure and temperature conditions, hazardous and reactive chemicals, ionizing and non-ionizing radiation, open flame and explosion risks, and biological hazards such as pathogenic microorganisms and their associated toxins.

Biosecurity:

Is the protection of individuals from pathogens by bad guys.

Biosafety:

Protects people from dangerous pathogens.

Objectives of the biosafety study:-

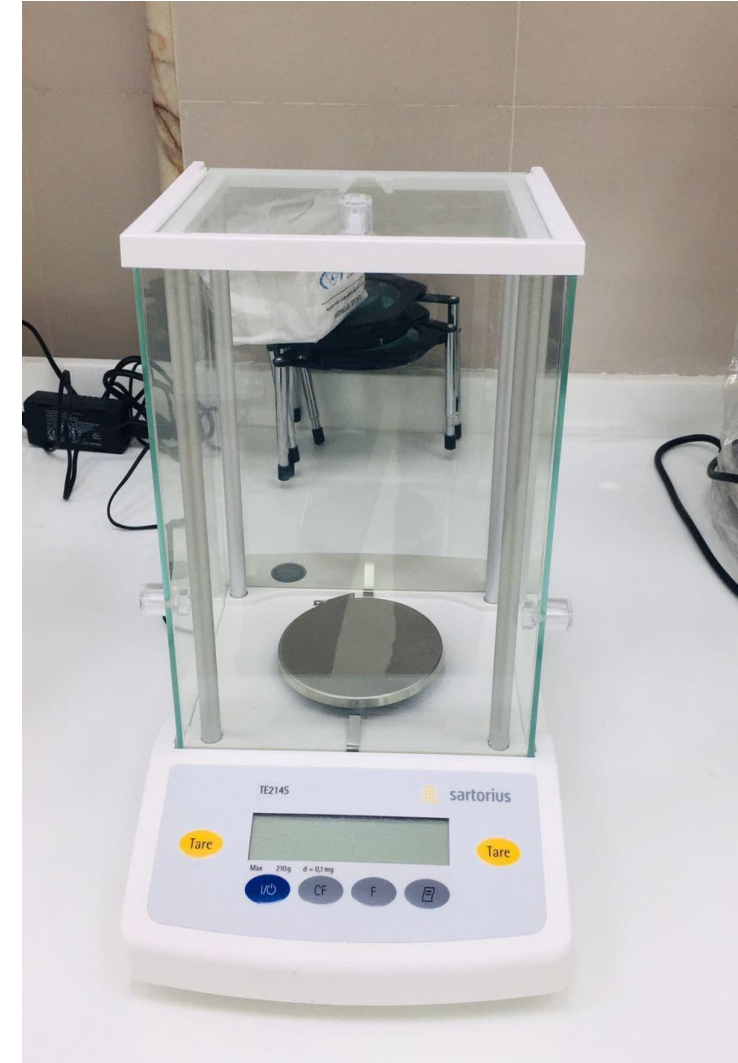
Reduces exposure of laboratory workers as well as people outside the laboratory and the surrounding environment to infectious agents.



Laboratory Equipment Names and Functions:

1- Balance:

An instrument for determining weight.



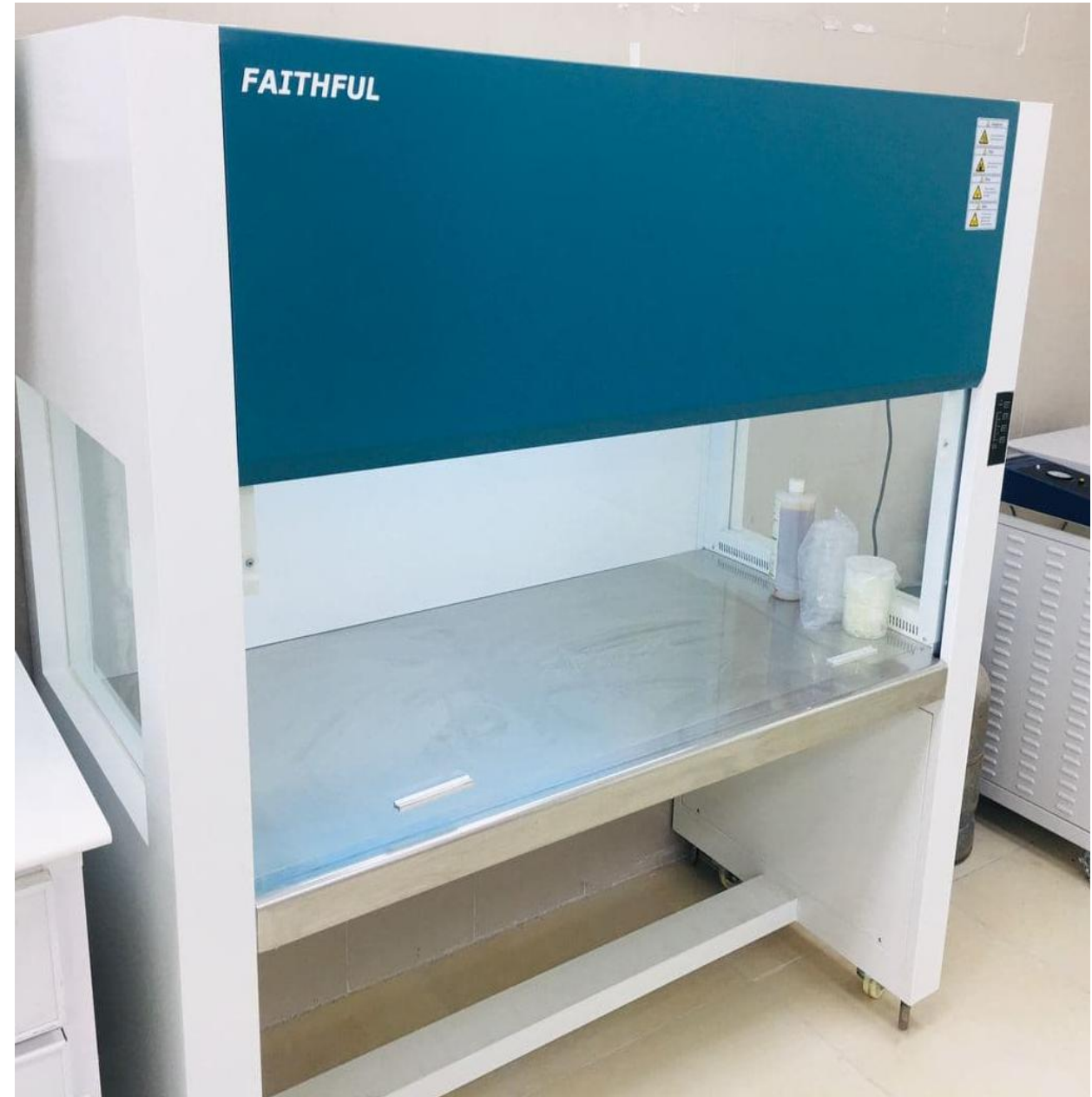
2-Autoclave

Is a pressure chamber used to **sterilize** equipment by subjecting them to high pressure saturated steam at 121 °C for around 15–20 minutes.



3-Fume Hood

Used to prevent a person from exposure to hazardous fumes from chemical reactions.



4-Water Distiller:

Is a water treatment method that produces contaminant-free water by convert water into vapor and returning it to a liquid state.



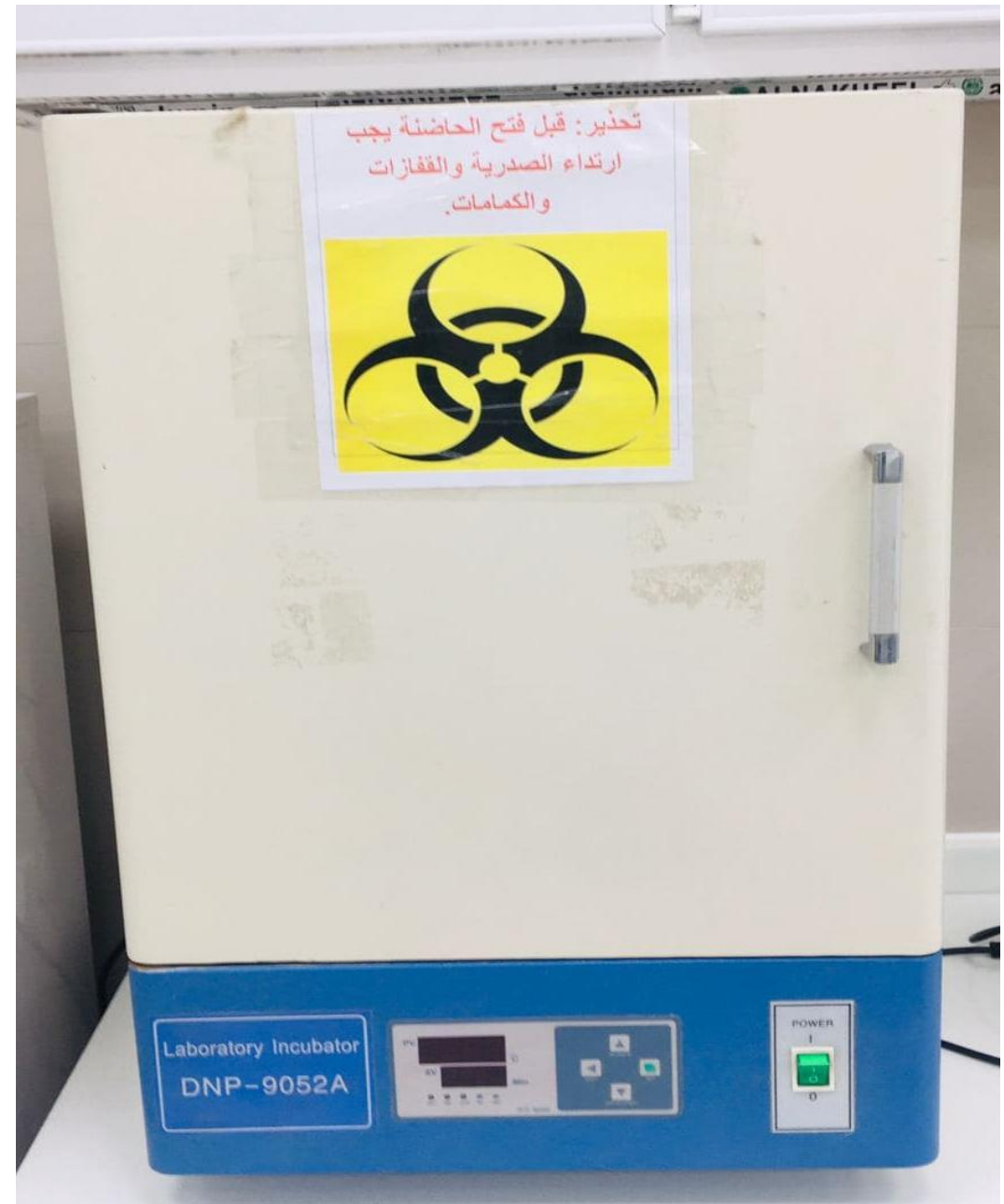
5-Hot Air Oven:

Is used for sterilization of glassware and materials that are spoiled by moist heat.



6-Incubator:

Is a device used to grow and maintain microbiological cultures .



7-Burner:

Used to heat substances and Sterilization.



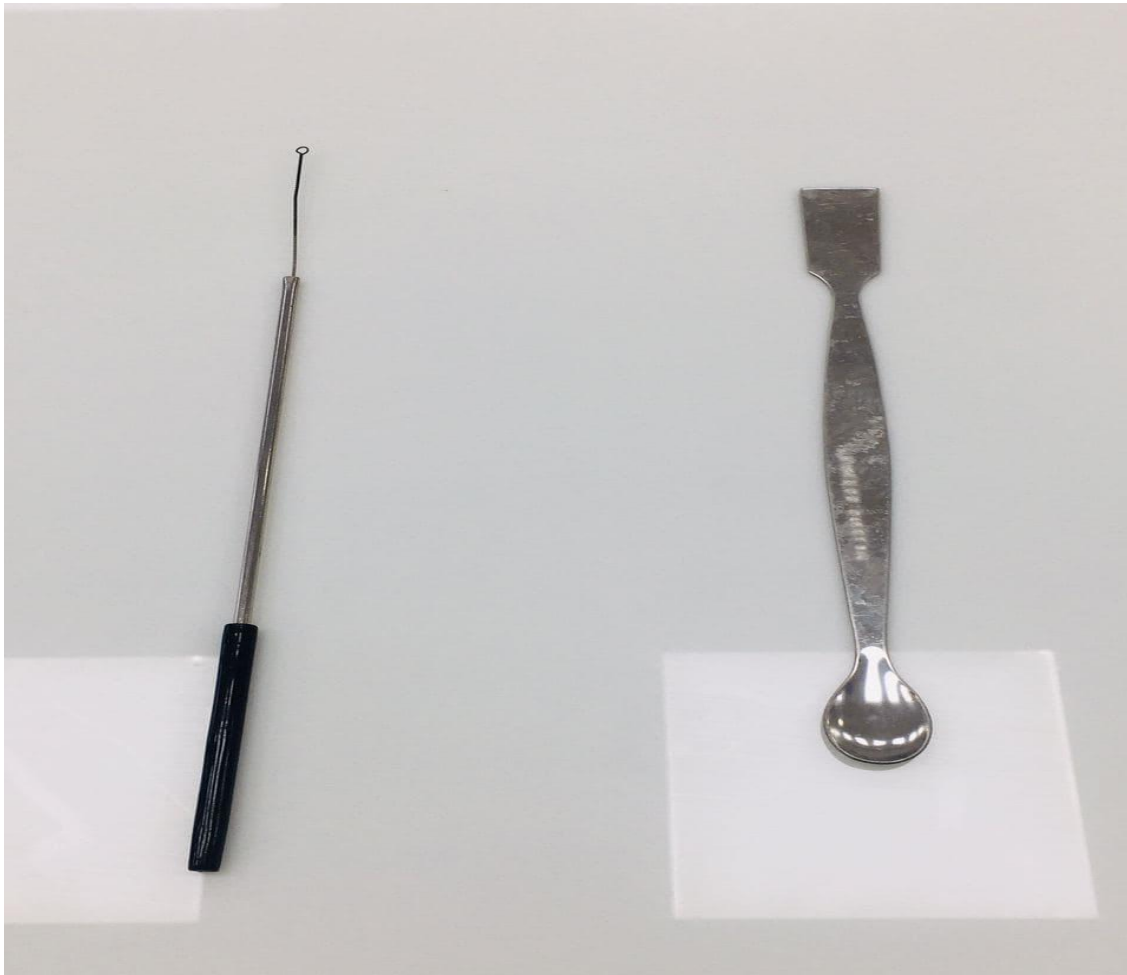
8- Conical Flask: used to heating and hold liquids.

9- Beaker: used to hold liquids.

10- Cylinder: used to measure of liquids.



Loop and Spatula



Filter paper



Personal and Population Protection

- 1- Wear lab coat in the proper manner.**
- 2- Wear gloves in the proper manner.**
- 3- Wear a face shield in the proper manner.**
- 4- Wear goggles in the proper manner.**



Before you leave the Laboratory:

- 1- Clean your work bench with disinfectant at the beginning and at the end of the laboratory .**
- 2- Leave all equipment , samples , and reagents in the laboratory.**
- 3- Leave your laboratory coat in the laboratory.**
- 4- Wash your hands with soap and water.**





Any Question?



Thank you